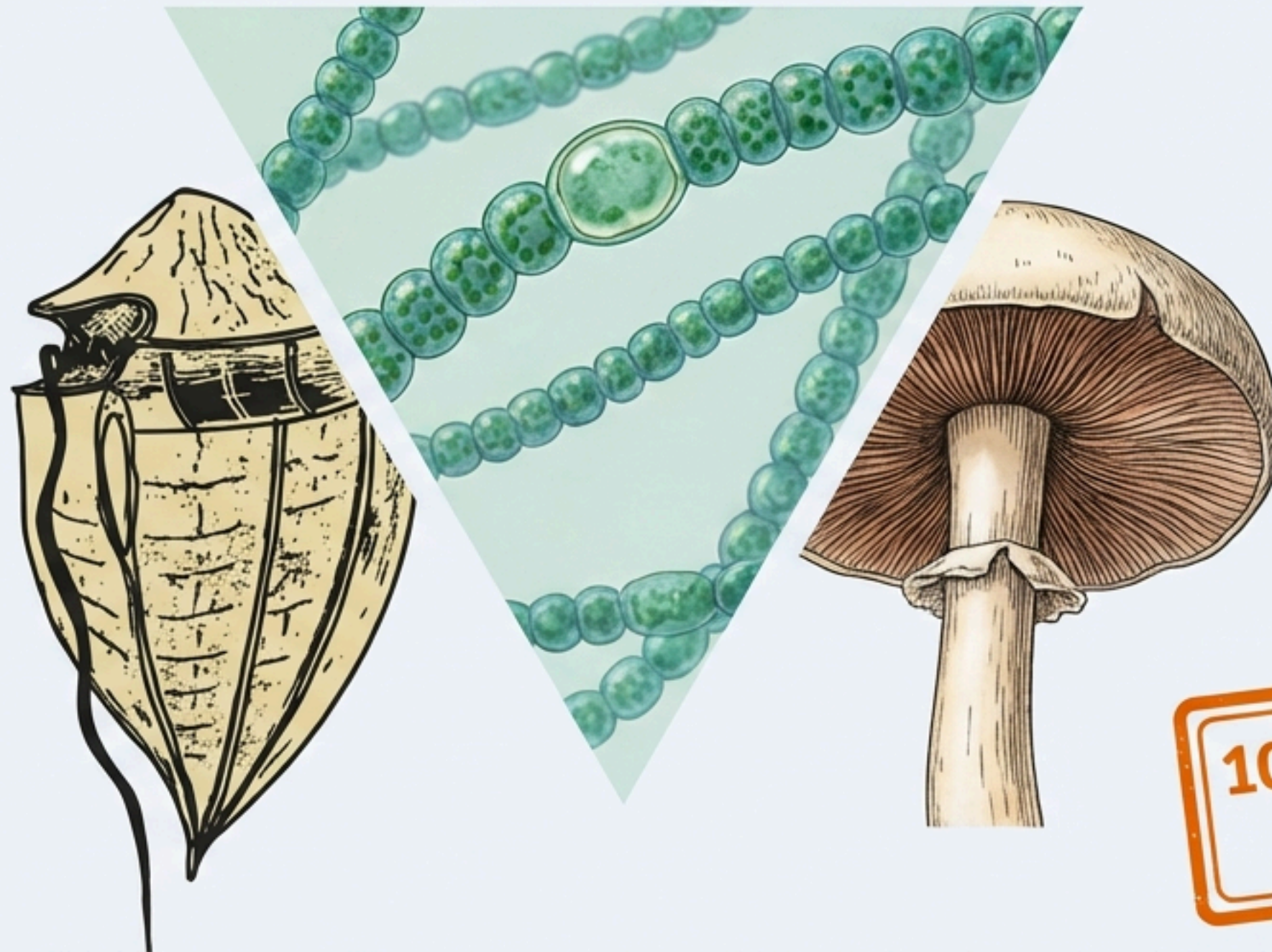


BIOLOGICAL CLASSIFICATION

NEET-UG Botany | Class 11 | Chapter 2



**100% NCERT Focused
| Exam-Oriented**

THE EVOLUTION OF CLASSIFICATION SYSTEMS

Note: Weightage: High
(3-4 Questions/Year)

Aristotle

Linnaeus

R.H. Whittaker (1969)

Scientific Basis: Morphology

 **Plants:** Trees, Shrubs, Herbs.

 **Animals:** Red blood (Eneima) vs. No red blood (Anaima).



 **Two Kingdom System:** Plantae & Animalia. 

 **Drawback:** Grouped Prokaryotes/Eukaryotes, Unicellular/Multicellular together.  

Five Kingdom Classification



Monera



Protista



Fungi



Plantae



Animalia

Whittaker's 5 Criteria (MANDATORY MEMORY)

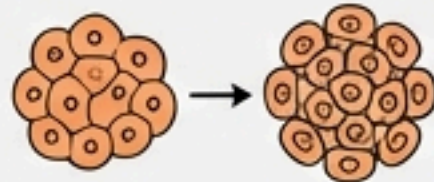
1



Prokaryotic vs Eukaryotic

Cell structure

2



Thallus

Body organization

3

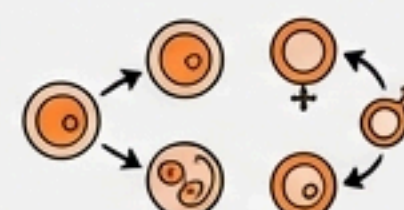


Auto

Heterotrophic

Mode of nutrition

4



Asexual

Sexual

Reproduction

5



Phylogenetic relationships

KINGDOM MONERA: ARCHAEABACTERIA & EUBACTERIA

Archaeobacteria (Harsh Habitats)

Key Feature: Different cell wall structure responsible for survival in extreme conditions.

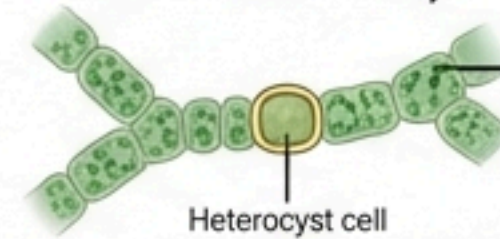
- **Halophiles:** Extreme salty areas. 
- **Thermoacidophiles:** Hot springs. 
- **Methanogens:** Marshy areas & gut of ruminants (produce methane/biogas). 

Eubacteria ('True Bacteria')

Rigid cell wall + flagellum (if motile). 

1. Photosynthetic Autotrophs (Cyanobacteria):

- Chlorophyll 'a' similar to green plants.
- Unicellular, Colonial, or Filamentous.



Heterocysts: Specialized cells for N_2 -fixation (e.g., Nostoc, Anabaena).

2. Chemosynthetic Autotrophs:

- Oxidise nitrates/nitrites for ATP. Recycle N, P, Fe, S.

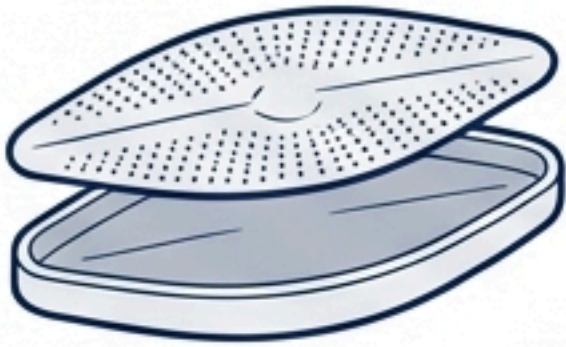
3. Heterotrophic:

- Decomposers (Curd, Antibiotics, Legume N-fixation).
- Pathogens: Cholera, Typhoid, Tetanus, Citrus canker. 

Mycoplasma:  No cell wall. Smallest living cells. Survive without oxygen. Pathogenic.

KINGDOM PROTISTA: THE BOUNDARIES ARE NOT WELL DEFINED

Chrysophytes (Diatoms)



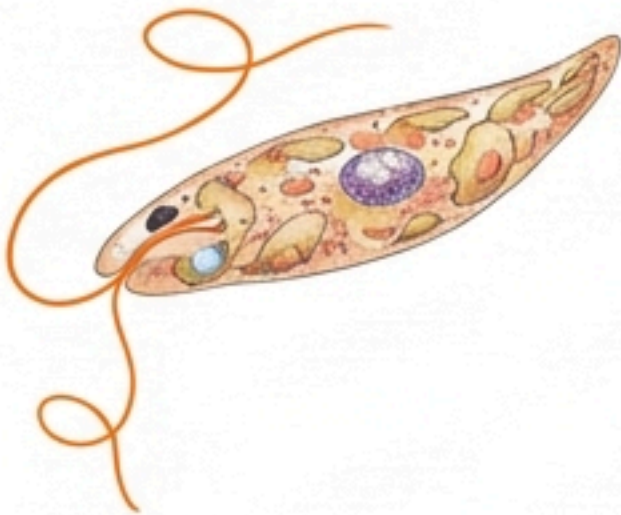
- **Habitat:** Fresh/Marine Plankton.
- **Cell Wall:** Two thin overlapping shells ('Soap box') embedded with Silica (Indestructible).
- **Diatomaceous Earth:** Polishing, filtration. Chief producers in oceans.

Dinoflagellates



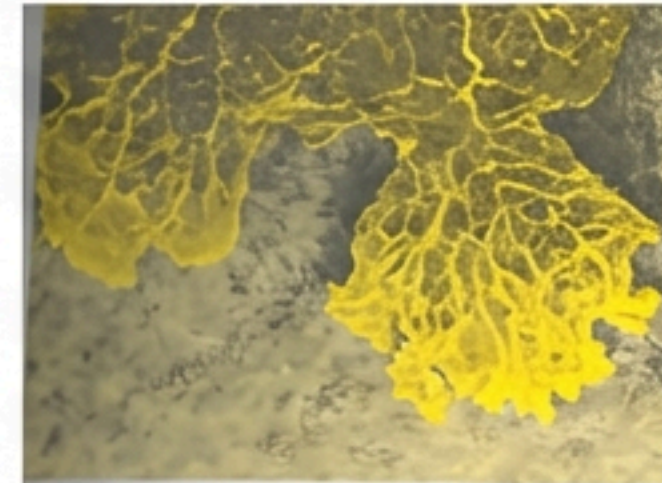
- Stiff cellulose plates on outer surface.
- 2 Flagella (longitudinal & transverse).
- **Red Tides:** Rapid multiplication of Gonyaulax.

Euglenoids



- No cell wall. Have Pellicle (protein rich) = Flexible body.
- **Mixotrophs:** Photosynthetic in light; Predators in dark.

Slime Moulds

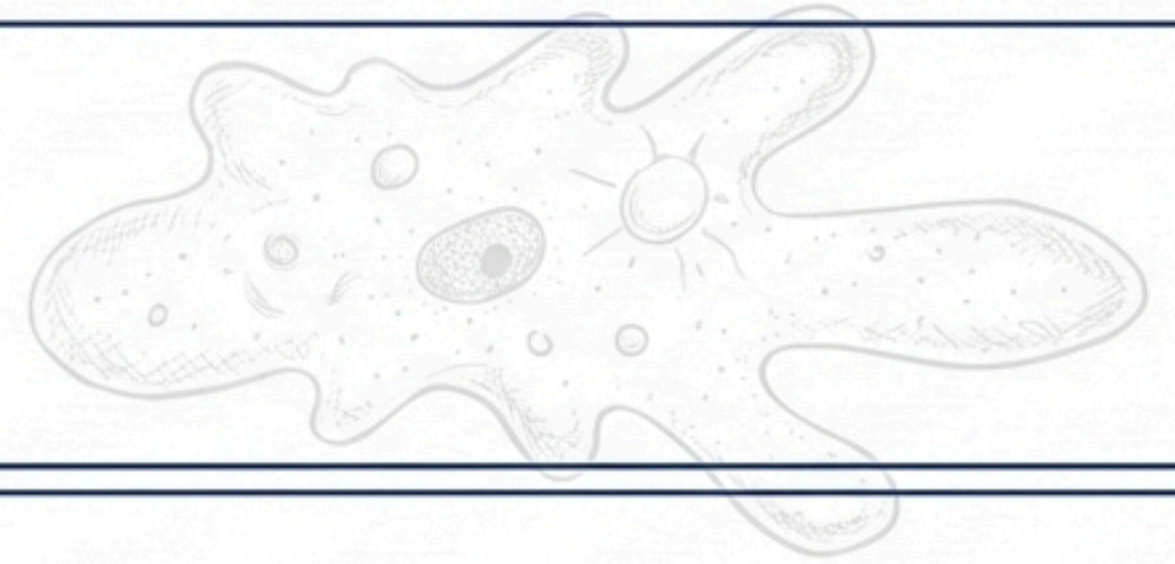


- Saprophytic.
- Plasmodium (aggregation) under suitable conditions.
- Spores formed in fruiting bodies (True walls, highly resistant).

KINGDOM PROTISTA: PROTOZOANS (HETEROTROPHS)

Amoeboid

- Move via **Pseudopodia** (false feet).
- Marine forms have silica shells.
- Ex: *Amoeba*, *Entamoeba* (Parasite).



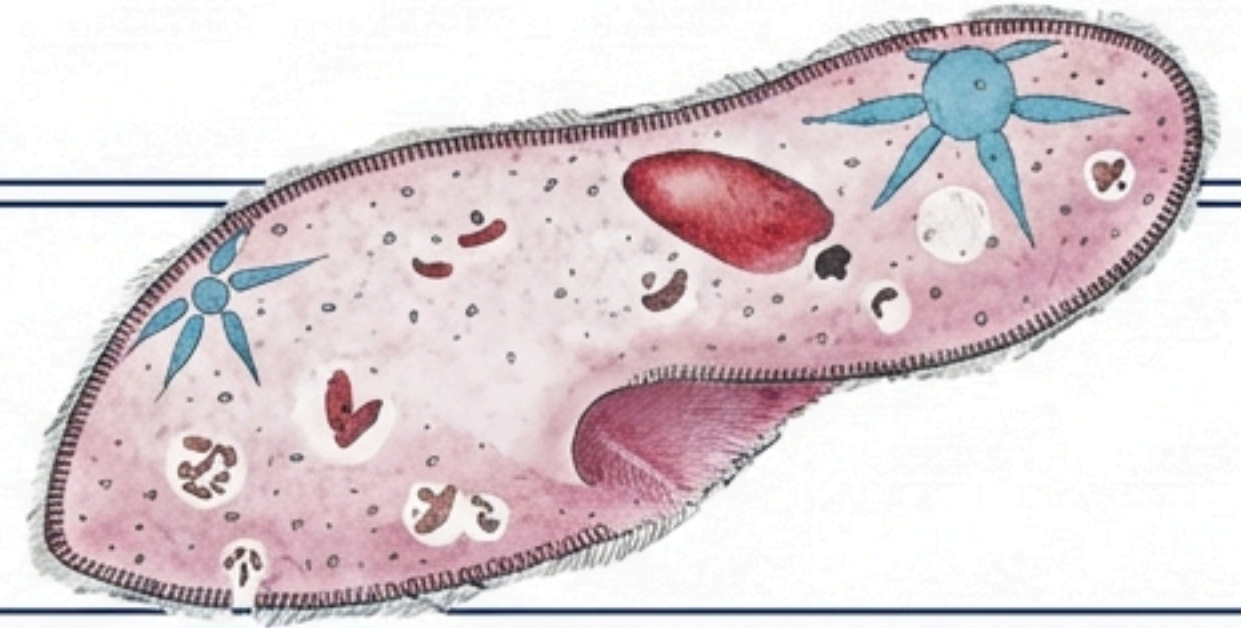
Flagellated



- Free-living or parasitic.
- Ex: *Trypanosoma* (Sleeping Sickness).

Ciliated

- Actively moving (thousands of cilia). Have a **Gullet**.
- Ex: *Paramecium*.



Sporozoans

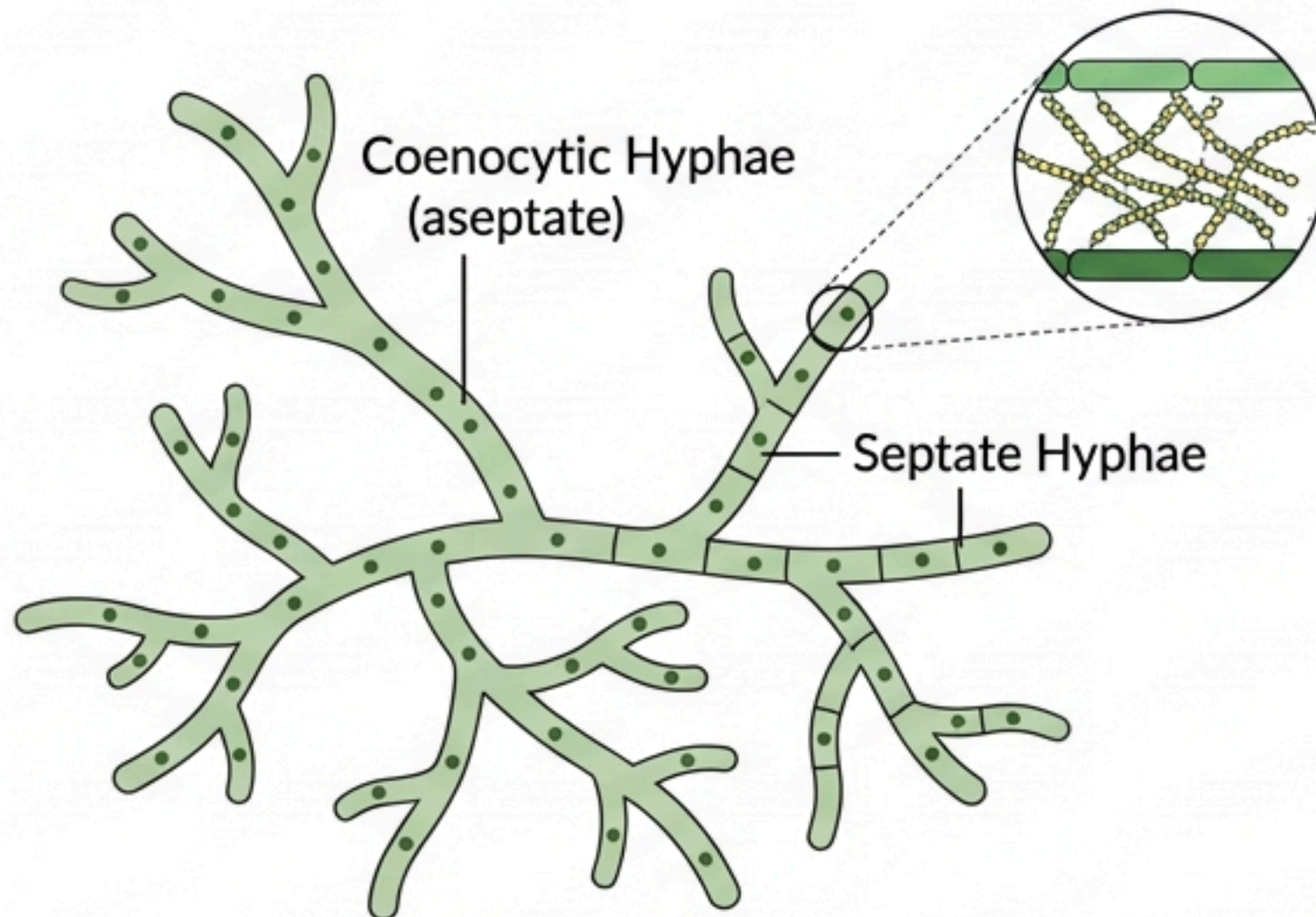


- Infectious spore-like stage.
- Ex: *Plasmodium* (Malarial parasite).

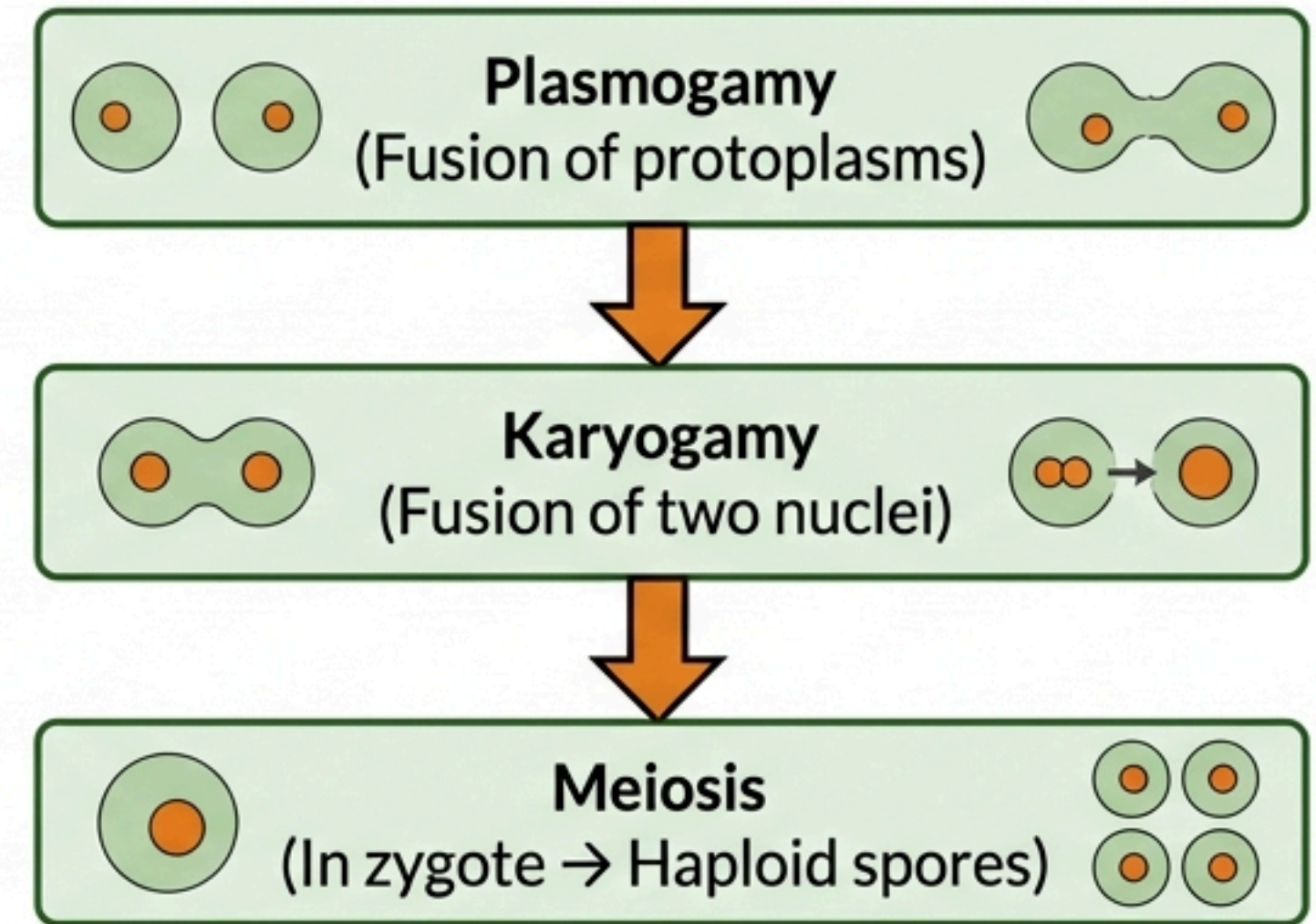
KINGDOM FUNGI: STRUCTURE & REPRODUCTION

General Structure

- **Filamentous** (Except Yeast - Unicellular).
- **Hyphae**: Thread-like structures → Mycelium (Network).
- **Coenocytic**: Continuous tubes, multinucleated (aseptate).
- **Septate**: Cross-walls present.
- **Cell Wall**: Chitin + Polysaccharides.



Sexual Cycle (3 Steps)



Concept: Dikaryophase ($n + n$)

In Ascomycetes & Basidiomycetes, fusion of nuclei is delayed. Cells have two nuclei (Dikaryon) before fusion.



Dikaryon ($n + n$)

COMPARATIVE CLASSIFICATION OF FUNGI CLASSES

	Phycomycetes	Ascomycetes (Sac)	Basidiomycetes (Club)	Deuteromycetes (Imperfect)
Mycelium	Aseptate, Coenocytic	Branched, Septate	Branched, Septate	Branched, Septate
Asexual Spores	Zoospores (motile)/ Aplanospores. (Endogenous)	Conidia (Exogenous)	Absent (Vegetative fragmentation)	Conidia only
Sexual Spores	Zygospore	Ascospores (Endogenous in Asci)	Basidiospores (Exogenous on Basidium)	Unknown / Absent
Examples & Visuals	Mucor, Rhizopus, Albugo	Penicillium, Yeast, Neurospora. 	Agaricus, Ustilago, Puccinia. 	Alternaria, Trichoderma

VIRUSES, VIROIDS, PRIONS & LICHENS (NON-WHITTAKER GROUPS)

VIRUSES

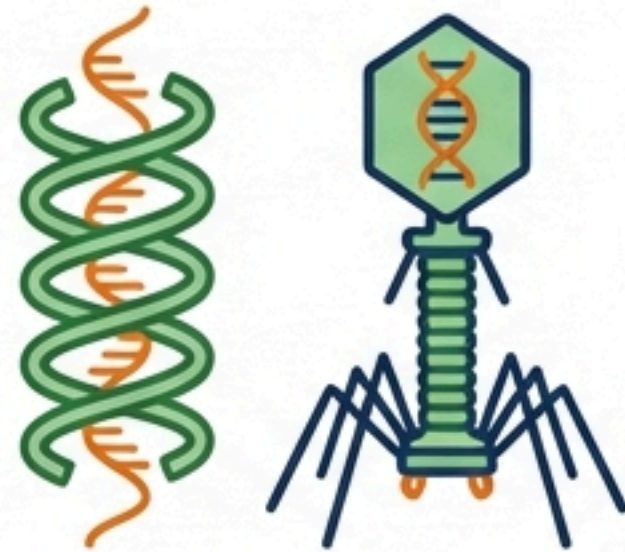
Non-cellular, inert crystal structure outside host. Obligate parasites.

Genetic Material: RNA OR DNA (Never both).

Protein Coat: Capsid made of Capsomeres.

Plant Viruses: ssRNA.

Bacteriophages: dsDNA.

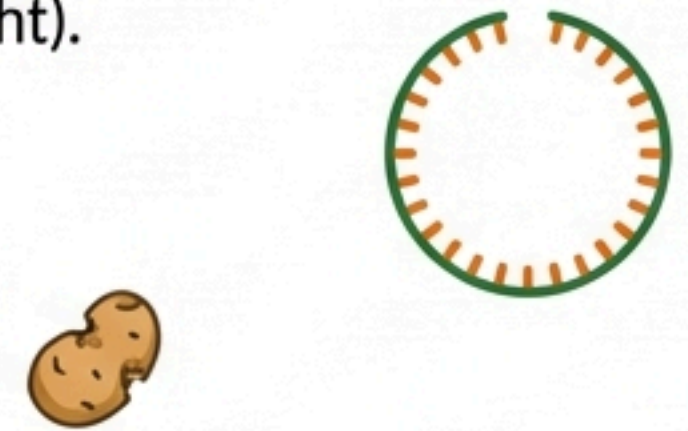


VIROIDS (T.O. Diener, 1971)

Free RNA (Low molecular weight).
No protein coat.

Smaller than viruses.

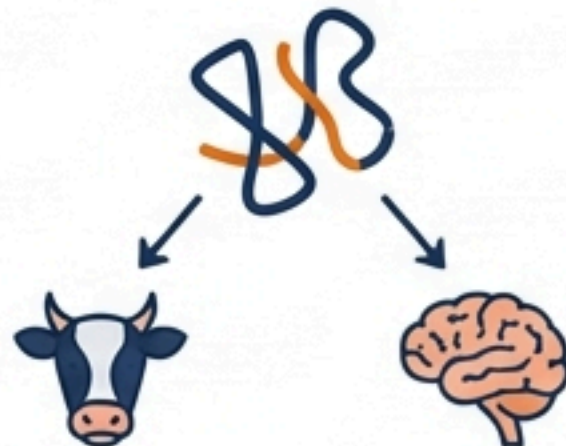
Disease: Potato Spindle Tuber disease.



PRIONS

Abnormally folded proteins.

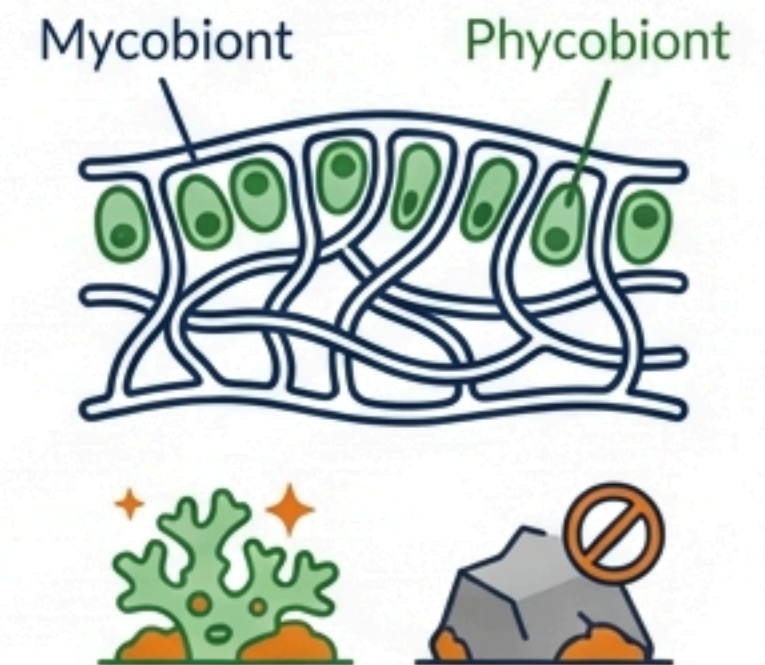
Diseases: BSE (Mad Cow), CJD (Humans).



LICHENS

Symbiosis: Phycobiont (Algae - Autotrophic) + Mycobiont (Fungi - Heterotrophic).

Pollution Indicators:
Do not grow in polluted areas.



⚠ NCERT LINE EMPHASIS (DIRECT NEET PICKUPS)

“The cell walls in *diatoms* are embedded with silica and thus the walls are **indestructible**.”



“*Mycoplasma*... are the **smallest living cells** known and can **survive without oxygen**.”



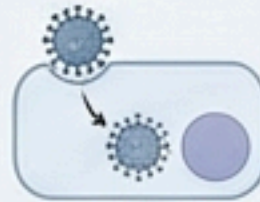
The fungi **filamentous**... with the exception of **yeasts which are unicellular**.”



No virus contains **both RNA and DNA**.”



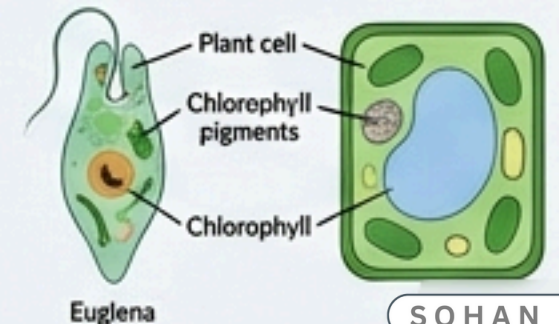
Viruses are **obligate parasites**.”



Members of Phycomycetes are found... as **obligate parasites on plants**.”

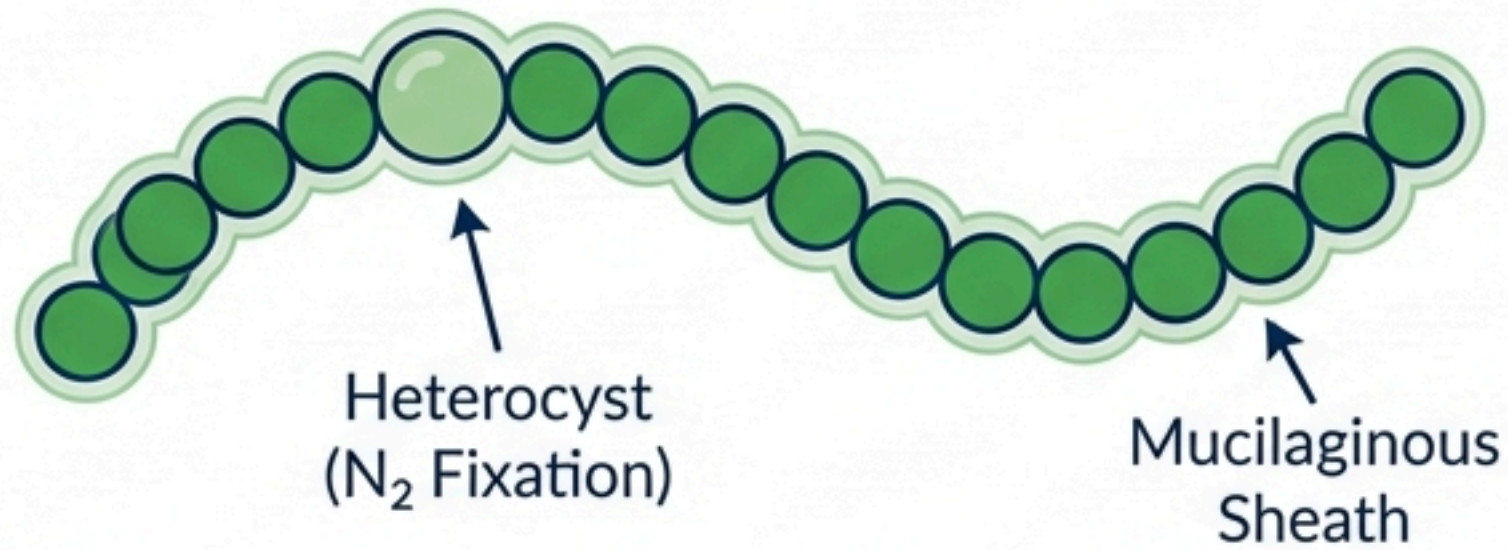


Interestingly, the pigments of **euglenoids** are **identical** to those present in higher plants.”

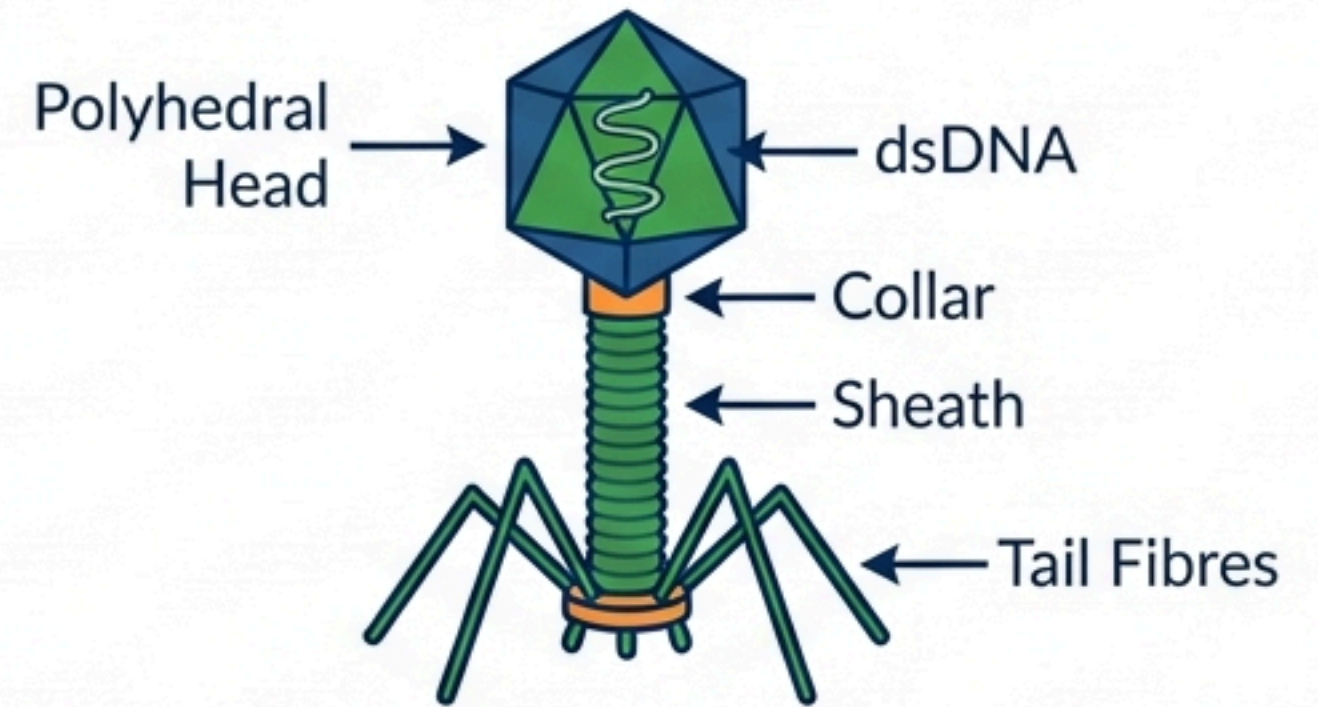


VISUAL FOCUS & DIAGRAM LABELING

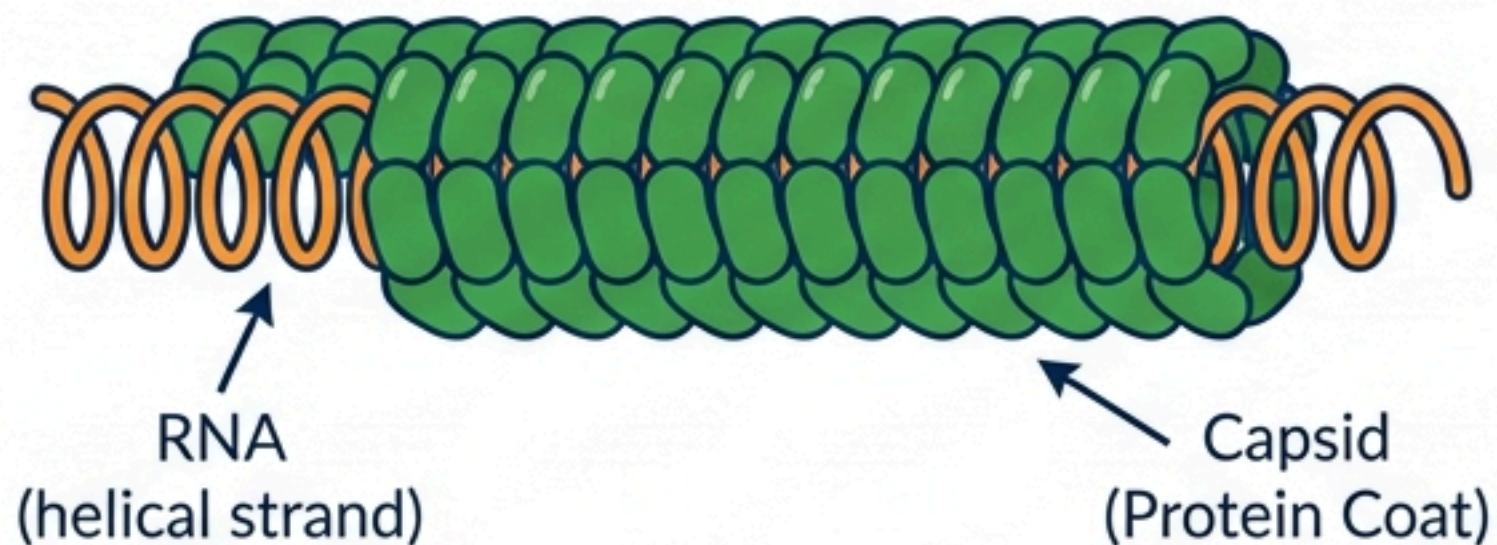
Nostoc (Cyanobacteria)



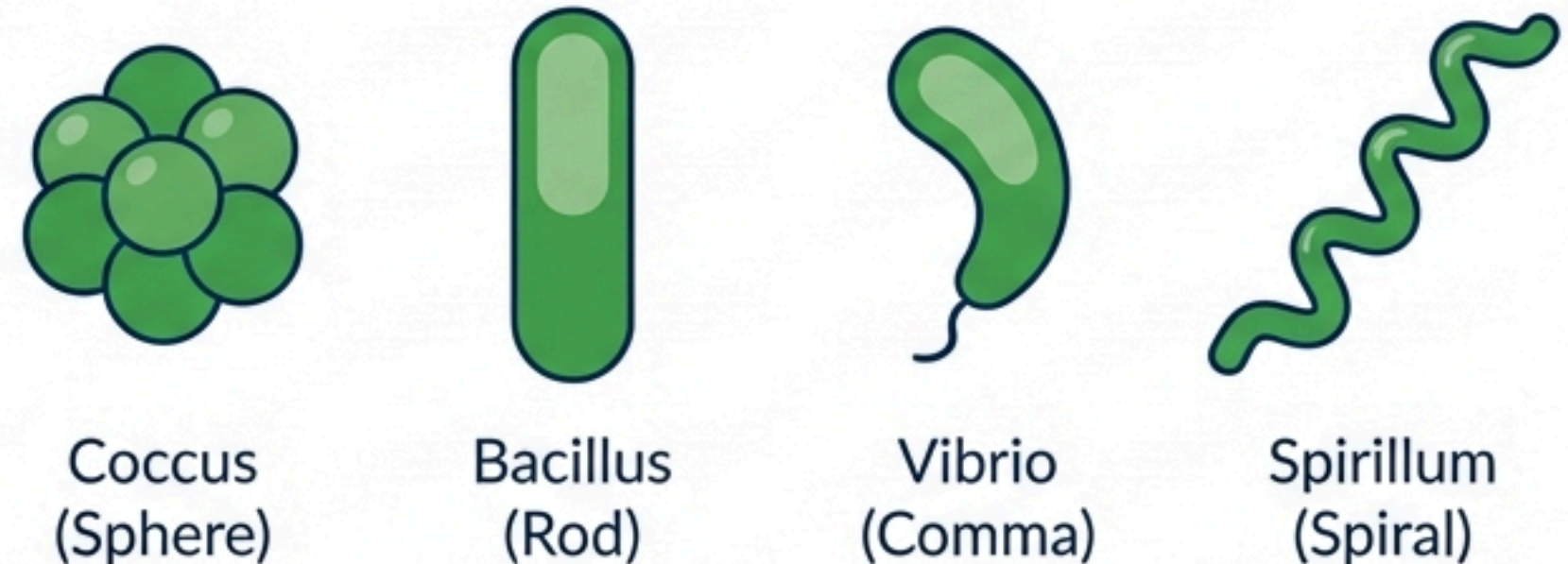
Bacteriophage



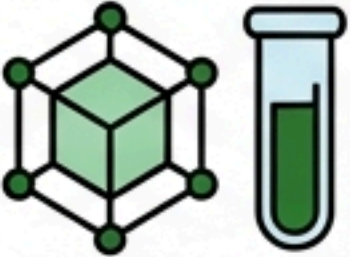
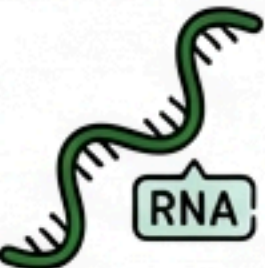
TMV (Tobacco Mosaic Virus)



Bacteria Shapes



HISTORY OF VIROLOGY & CLASSIFICATION

	Dmitri Ivanowsky (1892)	• Recognized microbes of Tobacco Mosaic Disease. • Smaller than bacteria (passed bacteria-proof filters).
	M.W. Beijerinck (1898)	• Demonstrated extract of infected plants causes infection. • Termed "Contagium vivum fluidum" (Infectious living fluid).
	W.M. Stanley (1935)	• Showed viruses could be crystallised. • Crystals consist largely of proteins.
	T.O. Diener (1971)	• Discovered Viroids (Free RNA).

CONFUSING PAIRS & EXAM TRAPS

Spore Origin (Endogenous vs (Endogenous vs Exogenous) 	'Red' Phenomena 	Parasitic Fungi 
Endogenous (Inside): Phycomycetes (Zoospores/Aplanospores), Ascomycetes (Ascospores). Exogenous (Outside): Ascomycetes (Conidia), Basidiomycetes (Basidiospores).	Red Tides: Gonyaulax (Dinoflagellates). Red Sea: Trichodesmium (Cyanobacteria).	Rust: Puccinia (Basidiomycetes). Smut: Ustilago (Basidiomycetes). White Spots on Mustard: Albugo (Phycomycetes).

CONCEPTUAL LOGIC CHECKS (ASSERTION-REASON)

CASE 1: NEUROSPORA	CASE 2: EUGLENOIDS	CASE 3: FUNGI DIKARYOTIC STAGE
		
Assertion: <i>Neurospora</i> is used extensively in biochemical and genetic work. Reason: It belongs to Basidiomycetes. Analysis: Assertion is True, Reason is False. (<i>Neurospora</i> is a an Ascomycete).	Assertion: Euglenoids behave like heterotrophs in the absence of sunlight. Reason: They lack photosynthetic pigments. Analysis: Assertion is True, Reason is False. (They HAVE pigments identical to higher plants).	Assertion: Fungi show a dikaryotic stage ($n+n$). Reason: In Ascomycetes and Basidiomycetes, karyogamy is delayed after plasmogamy. Analysis: Both True & Correct Explanation.

RAPID REVISION KEYWORDS



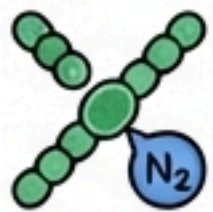
- Siliceous wall / Soap box → **Diatoms (Chrysophytes)**



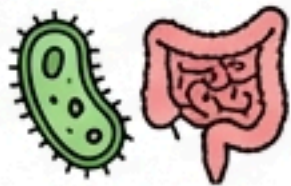
- Pellicle / Flexible → **Euglenoids**



- Stiff Cellulose Plates → **Dinoflagellates**



- Heterocyst → **N₂ Fixation (Nostoc/Anabaena)**



- Methanogens → **Archaeobacteria** (Gut of cows)



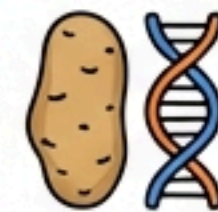
- Coenocytic Mycelium → **Phycomycetes**



- Dikaryon (n+n) → **Ascomycetes & Basidiomycetes**



- Algal Bloom → **Cyanobacteria** (Polluted water)



- **Viroid** → Free RNA, **Potato Spindle Tuber disease**



- **Prion** → Abnormally folded protein, **Mad Cow disease**